

ADI HADITYA NURSYAM

SOFTWARE, AUTOMATION & MECHANICAL ENGINEERING

+62 858-7714-3492 | soezyxst@gmail.com | Bandung, Indonesia
linkedin.com/in/adihnursyam | github.com/soezyxstt | adihnursyam.com

PROFESSIONAL SUMMARY

ITB Mechanical Engineering graduate (GPA 3.90/4.00), education founder, and technology product builder with experience spanning production software, AI, automation, robotics, mechanical analysis, teaching, and organizational leadership. Experienced in taking problems from definition through design, implementation, validation, and operation.

CORE COMPETENCIES

Software: JavaScript, TypeScript, React, Next.js, Node.js, APIs, authentication, RBAC, testing, Git

Data, Cloud & AI: PostgreSQL, SQLite/Turso, Drizzle, Prisma, Cloudflare Workers/D1/R2/Vectorize, Docker, Gemini, RAG

Automation & Robotics: ESP32, C/C++, PID, telemetry, Web Serial, MQTT, industrial IoT, Python, MATLAB

Mechanical: SOLIDWORKS, part and assembly CAD, prototype integration, ANSYS Workbench, FEA, maintenance and failure analysis

Leadership: product ownership, technical leadership, teaching, program delivery, cross-functional coordination

EDUCATION & RECOGNITION

Bachelor of Engineering, Mechanical Engineering | Institut Teknologi Bandung **Aug 2022 - Jul 2026**

- GPA 3.90/4.00 | Mechanical Engineering Outstanding Student 2022 (one of three departmental students).

SOFTWARE & PRODUCT EXPERIENCE

Founder & Lead Full-Stack Engineer (CEO) | Zyx Academy (zyxacademy.com)

2024 - Present

Bandung, Indonesia

- Lead strategy, product, engineering, learning operations, and tutoring for an active AI-enabled education initiative.
- Built and operate the production platform as primary engineer, owning frontend, backend, relational/vector data, cloud infrastructure, and content-authoring workflows.
- Serve 30+ learners with 100+ learning materials, assessments, spaced-repetition flashcards, progress tracking, analytics, and Zyra, a retrieval-backed AI tutor.

Head of IT Development / Full-Stack Engineer | HMM ITB

Aug 2024 - May 2026

Bandung, Indonesia

- Independently built and operated a typed learning and community platform supporting courses, assessments, events, discussions, forms, content, and administration.
- Delivered authentication, role-based access, relational data, storage, notifications, PWA capabilities, migrations, and tests using Next.js, TypeScript, tRPC, PostgreSQL, Prisma, Auth.js, Zod, and Vitest.

Head of Information Technology | PEMIRA KM ITB

Nov 2023 - Apr 2024

Bandung, Indonesia

- Led technical delivery for a team-built university election platform with authentication, typed APIs, relational data, administration, visualization, and Docker deployment.

ENGINEERING EXPERIENCE

Mechanical Maintenance Intern | PLN Indonesia Power

Jun 2025 - Aug 2025

Cilegon, Banten

- Participated in field inspection and maintenance of coal-handling and auxiliary equipment at Banten I Suralaya Unit 8.
- Investigated recurring BC-54 conveyor-idler failures using measurements, operating data, 3D modeling, and ANSYS Workbench FEA at an 800 TPH condition.
- Calculated 68.77 MPa maximum von Mises stress and a 1.74 minimum safety factor; proposed periodic ultrasonic thickness inspection after identifying abrasive wear plus fatigue as the more plausible mechanism.

Mechanical Engineer (Organizational/Volunteer) | Unit Robotika ITB

Jun 2022 - Jul 2025

Bandung, Indonesia

- Designed robot components and assemblies in SOLIDWORKS and coordinated interfaces across mechanical, electrical, and software workstreams.
- Led Sekolah URO's internal technical-training program, coordinating instructors and delivery for member development.

SELECTED PROJECTS

Two-DOF SCARA Robot

2026

- Built solo across SOLIDWORKS mechanical design, 3D-print-ready components, electromechanical assembly, ESP32 control, PID/feedforward, telemetry, Next.js/Web Serial HMI, and Python analysis for five experiment families.

Me-Bot Personal Assistant

2026

- Built solo as a deployed local-first product with IndexedDB offline capture and sync, Next.js APIs, D1/R2/Vectorize, Gemini triage, semantic memory, Google integrations, and encrypted vault workflows.

IIoT Conveyor Interface

2025 - 2026

- Built a browser HMI connecting MQTT machine data, operator controls, diagnostics, and persisted application state.

TEACHING & LEADERSHIP

Academic Tutor (Part-Time) | Mechanical Engineering ITB / Orpheus Engine

Aug 2023 - Aug 2026

Bandung, Indonesia

- Provided structured concept review and problem-solving support; personally tutored 50+ students across teaching roles.

President | Forum Mahasiswa Garut ITB

Jan 2025 - Aug 2025

Bandung / Garut

- Led organizational renewal after five years of dormancy and rebuilt governance, member programs, documentation, and stakeholder relationships.

Chairperson | Gumelar

Feb 2026 - May 2026

Garut, Indonesia

- Directed collaboration across 10+ universities and delivered programs involving 30 children, 14 student volunteers, and a 94-participant gathering.